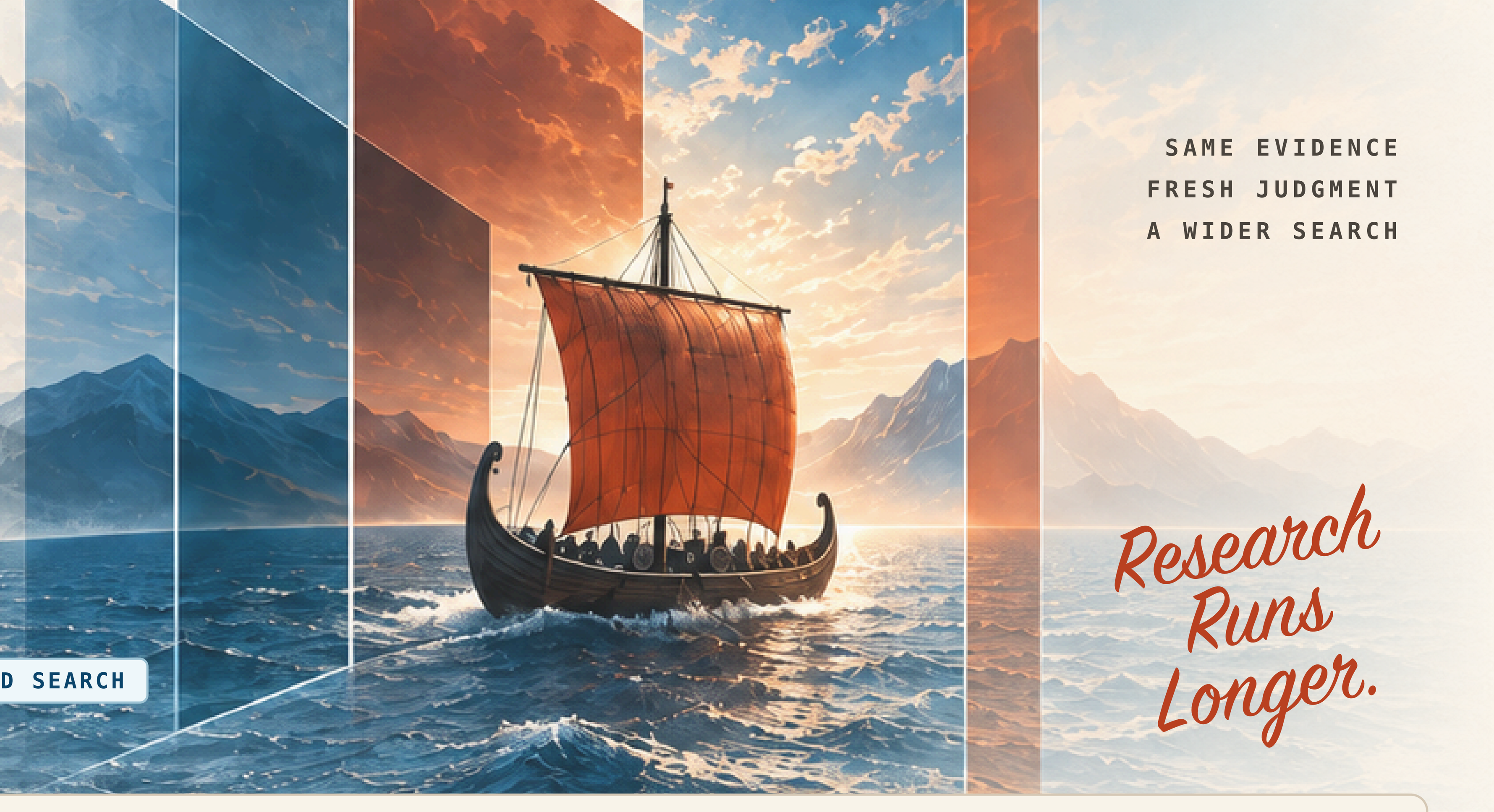
SciOdyssey.



A research layer over your agent harness.

— PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST · SUPERVISABLE · BROAD SEARCH

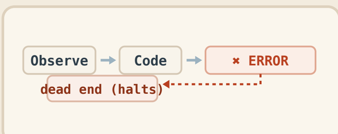
01 / PROBLEM

Why current agents fail.

Sustained autonomous research breaks in three different ways.

01 Closed-world fragility

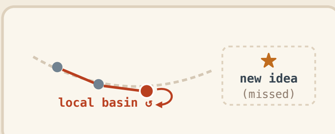
Fixed workflows shatter on unexpected runtime errors.



Unhandled runtime faults halt unassisted runs

02 Trajectory momentum

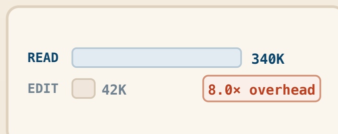
A single context carries past momentum into local basins.



22 runs trapped tweaking parameters in same basin

03 Context inflation cost

Reconstructing context costs far more than the edit itself.



10 files read to change 1 line; burns token budget

DESIGN RESPONSE Keep the world. Reset the trajectory.

02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

DUAL-LIFETIME LOOP — Persistent world · ephemeral trajectory · audited write-back

INPUTS: TASK, DATA, EVALUATOR, BUDGET

Fresh Scientist

Owns scientific judgment inside one trajectory: hypotheses, code, experiments, interpretation, and pivots.

→

META Review

Audits claims against artifacts, scopes conclusions, and decides what is durable enough to survive.

SELECTIVE PERSISTENCE Only audited empirical findings, code state, and bounded priors write back to the shared world.

PERSISTENT RESEARCH WORLD — What survives across trajectories

A · EVIDENCE
Knowledge + Ledger
Metrics, failures, reports, and experiment receipts in `ledger.jsonl`.

B · PRIORS
Brief + Hypotheses
Current understanding and exploration leads in `brief.md`, open to revision.

C · STATE
Code + State
Retained implementation, serialized model weights, checks, and git provenance.

CONTEXT RESET The next Scientist reads verified experience from disk, but does not inherit the previous reasoning trace.

RUNTIME SUBSTRATE

- **WORKTREE ISOLATION** Independent git worktrees isolate branches and preserve reproducible artifacts.
- **SELF-HEALING** Traceback → patch → rerun keeps unattended research moving after runtime faults.
- **PARALLEL SYNTHESIS** Completed branches can be compared post-hoc; useful ideas survive even when a branch loses.

> The next Scientist inherits **verified experience** — not **prior reasoning momentum**.
>
> Decoupled Memory · Audited Persistence · Fresh Scientific Judgment

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT — The model got better.

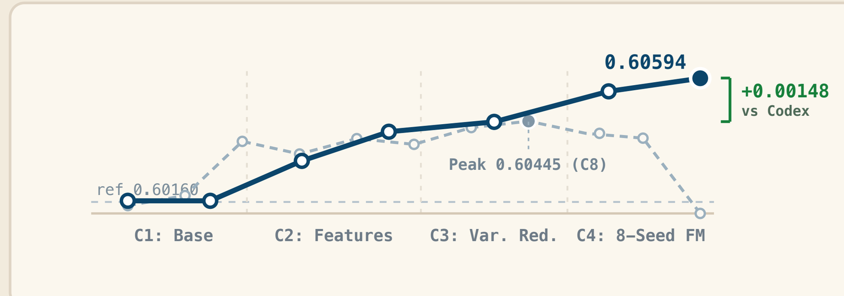
0.6016000 Official FM baseline → **0.6059363** SciOdyssey submission

+0.0043363 Primary improvement

GAUC	nDCG@5	LogLoss
0.6728421 (+0.0054)	0.5390304 (+0.0033)	0.38102 (-0.0032)

Feature Engine: 46 categorical cross-interactions + log1p transforms
Architecture: 8-seed Factorization Machine ensemble (NumPy / CPU)
Monotonic gain from baseline (0.6016) → feature engineering (0.6041) → seed ensembling (0.6059); **100% CPU inference** without GPU clusters.

02 SUSTAINED SEARCH — The search stayed productive.



4 cycles · 7 Full evaluations · E003–E013 frontier · +0.00148 over Codex

03 SELF-HEALING & GATEKEEPING — The agent recovered from failure.

TIER 1 · RUNTIME SELF-HEALING — CRASH AUTO-RECOVERY
FAILURE: Cycle 1 `numpy.float32` serialization crash
ACTION: Scientist parsed traceback, auto-patched scalar helper
OUTCOME: **0 human bugfixes** · Unattended rerun succeeded

TIER 2 · META GATEKEEPING — SPURIOUS STATE REJECTION
FALSE CLAIM: Scientist claimed BPR gain on medium proxy
ACTION: META audited `system/data.py`, caught `UNK` leak
OUTCOME: **Promotion blocked** · Zero contaminated state

Empirical receipts preserved: runtime self-healing telemetry and independent code-level proxy audits.

04 ARCHITECTURE ABLATION — Meta-scientist beats direct agents.

01 · DIRECT	02 · SUBAGENTS	03 · META-SCIENTIST
Antigravity	+ Subagents	Persistent World
0.60458	0.60472	0.60520
Single 2h run	Delegated swarm	+ fresh Scientist

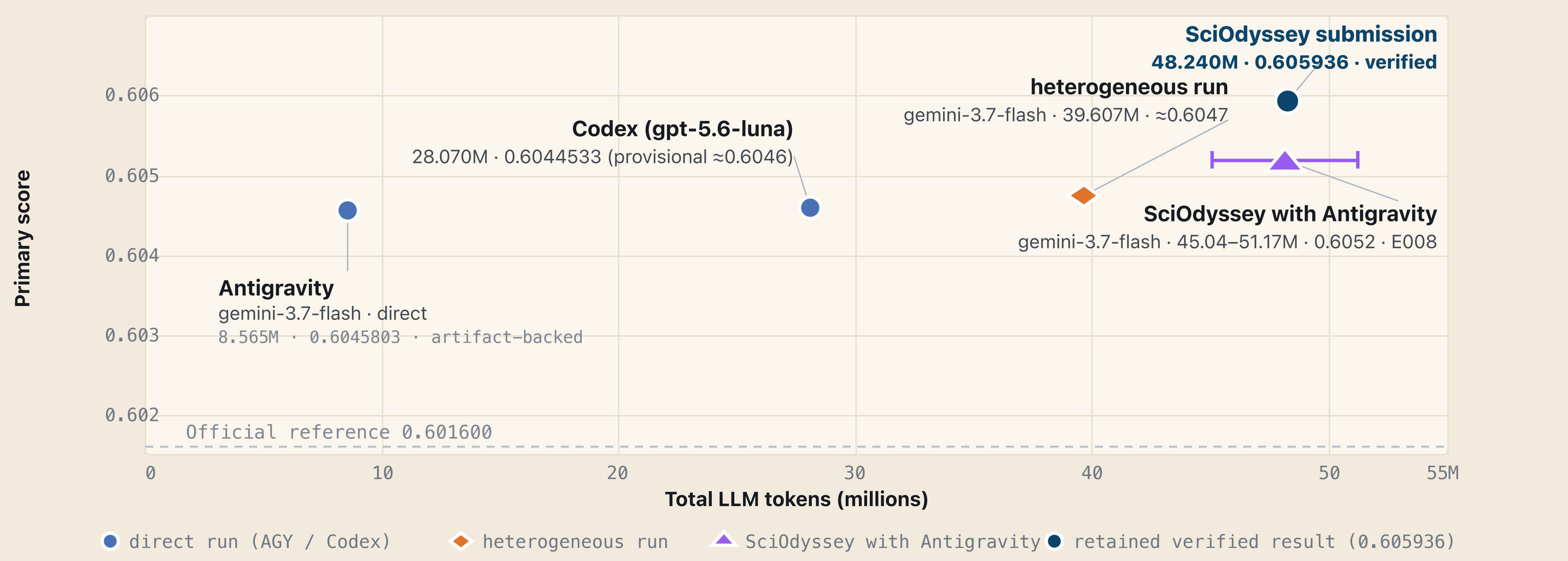
+0.00062 organization gain (Gemini 3.7 Flash)

Keep Gemini 3.7 Flash fixed and change only research organization: **direct agent (0.6046) → subagents (0.6047) → Meta-Scientist (0.6052)**.

05 RESOURCE ACCOUNTING — We are not tokenmaxxing. Score vs token investment.

~$10 cost · 1 × M2 · 0 GPU-hr · 1h 55m

KuaiRand-Pure · Public validation · Inclusive input+output tokens

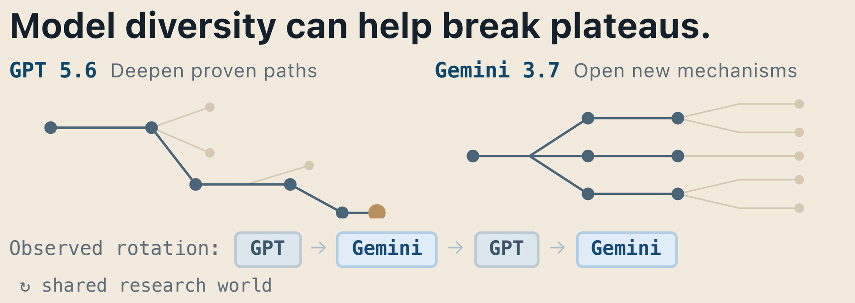


All results **verified on KuaiRand-Pure official validation split** with **complete reproducible artifacts**.

04 / FINDINGS

What the run revealed.

01 · MODEL DIVERSITY — Model diversity can help break plateaus.



Gemini-only **0.6052** → GPT+Gemini **0.6059363**

02 · PARALLEL SYNTHESIS — A losing branch can still improve the winner.

- A 0.6048 Redundant
- B 0.6048 Chosen ✓
- C 0.6039 Useful ideas

→ Reviewer (Synthesize B+C) → Fresh Scientist 0.6055536

Selection keeps useful knowledge, not just the top score: Branch C contributed pairwise ranking and rank normalization.

Branch B+C Ideas → Synthesis **0.6055536**

05 / EXPERIENCE

Start your journey.

Autonomous by default.
Supervisable by design.

task.md — What to solve

Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation protocol.

- Dataset paths & split integrity
- Official sandbox evaluator

PERSONAL.md — How to work

User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

- Model rotation (GPT / Gemini)
- Supervision & budget thresholds

SUPERVISION TIERS — **Tier 1** Unattended Overnight Runs · **Tier 2** Checkpoint Review (META) · **Tier 3** Steer Working Priors

1. Define task.md → 2. Single-step test → 3. Autonomous search → 4. Parallel explore → 5. Inspect evidence

DECLARATIVE PROTOCOL **Human defines boundary and preferences** (task.md + PERSONAL.md); **SciOdyssey autonomously iterates, heals, and crystallizes verified evidence.**

Coding agent harness · Execution studio — Works with: Codex · Claude Code · Antigravity · OpenCode · Trae

```
> SKILL.md "Use the research-agent skill. Explore in parallel; let me review what to keep."

$ research-agent step                              # one cycle · human reviews
  META → Scientist → Evidence → Your turn ✓

$ research-agent run --max-cycles 4                # autonomous · fresh scientist each round
  C1 ↺ → C2 ↺ → C3 ↺ → C4 · Candidate Retained ✓

$ research-agent parallel --branches 3             # isolated worktrees · reviewer synthesizes
  Branch A·B·C → Reviewer Synthesis → Explicit Promotion ✓

$ research-agent gui                               # dashboard · telemetry & checkpoints
  Telemetry · Loss Curves · Verified Checkpoints ✓
```

github.com/zc6600/research-agent-kuairand ↗
Team Good4AI: Chen Zhu · Zhou Ziyu · Shilin Xu · GE GAO · Jiran Li
Sources: slides/slides.md · docs/ARCHITECTURE.md · docs/FINAL_REPORT.md · KuaiRand-Pure submission ledger

PROJECT WIKI & DEMO
sciodyssey.zc6600.wiki ↗
Scan for live report & artifacts